

# Cooper Huntington-Bugg

Jenks, Oklahoma | cooperehuntington@gmail.com | github.com/Cooper-Bugg

## PROFESSIONAL SUMMARY

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Computer Science student at Oklahoma State University with a strong foundation in C++, embedded systems, and algorithmic problem solving. Experienced in developing high-performance, low-level systems and simulations with an emphasis on efficient memory and resource management. Seeking software engineering roles in C++ and embedded systems.

## TECHNICAL SKILLS

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**Languages:** C++, Java, Python, JavaScript, SQL, Assembly

**Core Concepts:** Embedded Programming, Memory Management, Bitwise Operations, Low-level I/O, Network Security

**Systems & Backend:** Event-Driven Architecture, Asynchronous Programming, REST APIs, Database Design

**Practices:** Git, Unit Testing, Debugging, Agile, CI/CD

## PROJECTS

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### The CPP Grind — C++, HTML, CSS, JavaScript

- Built a personal C++ learning site from the ground up, designed the frontend, wired the data layer and implemented localStorage-based progress tracking, driven by a passion for learning and teaching the language to others.
- Developed 43 hand-typed C++ projects spanning the full learning arc: CLI tools, POSIX pipelines, recursive backtracking solvers, concurrency primitives (mutexes, condition variables, lock-free queues), embedded simulators and a foundational systems library built from scratch.
- Created daily interactive challenges including a C++ keyword Wordle and code challenge engine (will this compile, buffer overflow detection, output prediction) to reinforce concept retention.

### Modular Discord Bot — Python, discord.py, asyncio

- Architected a modular, cog-based Discord bot with 60+ commands, separating command handlers from core logic and API integrations to support hot reloading of components at runtime.
- Developed asynchronous, event-driven systems using asyncio and aiohttp, integrating multiple external APIs while managing concurrent multiplayer state and implementing runtime monitoring features including latency tracking and error reporting.

## PROFESSIONAL EXPERIENCE

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### Huntington ID | Web Developer

January 2020 – Present

- Maintain and update a live WordPress site, implementing SEO improvements, plugin updates, and content revisions to ensure reliability and performance.

### Royale Landscaping | Landscaping Foreman

May 2016 – January 2020

- Led small teams to complete client projects on schedule, adapting plans under tight deadlines to ensure quality and client satisfaction.

## ACADEMIC PROJECTS

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### Capstone & Software Engineering Projects — Oklahoma State University

- Built and deployed 2 full-stack applications for Card Clash and Smart Scheduler, designing and wiring both frontends, REST API integrations, and backend infrastructure using Node.js and MySQL with asynchronous logic and persistent data storage.

## EDUCATION

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### Oklahoma State University

GPA 4.0

B.S. Computer Science | Minor in Mathematics | Expected December 2026

Undergraduate Certificate in Cybersecurity

### Tulsa Community College

GPA 4.0

A.S. Computer Information Systems | A.A. Liberal Arts | 2024